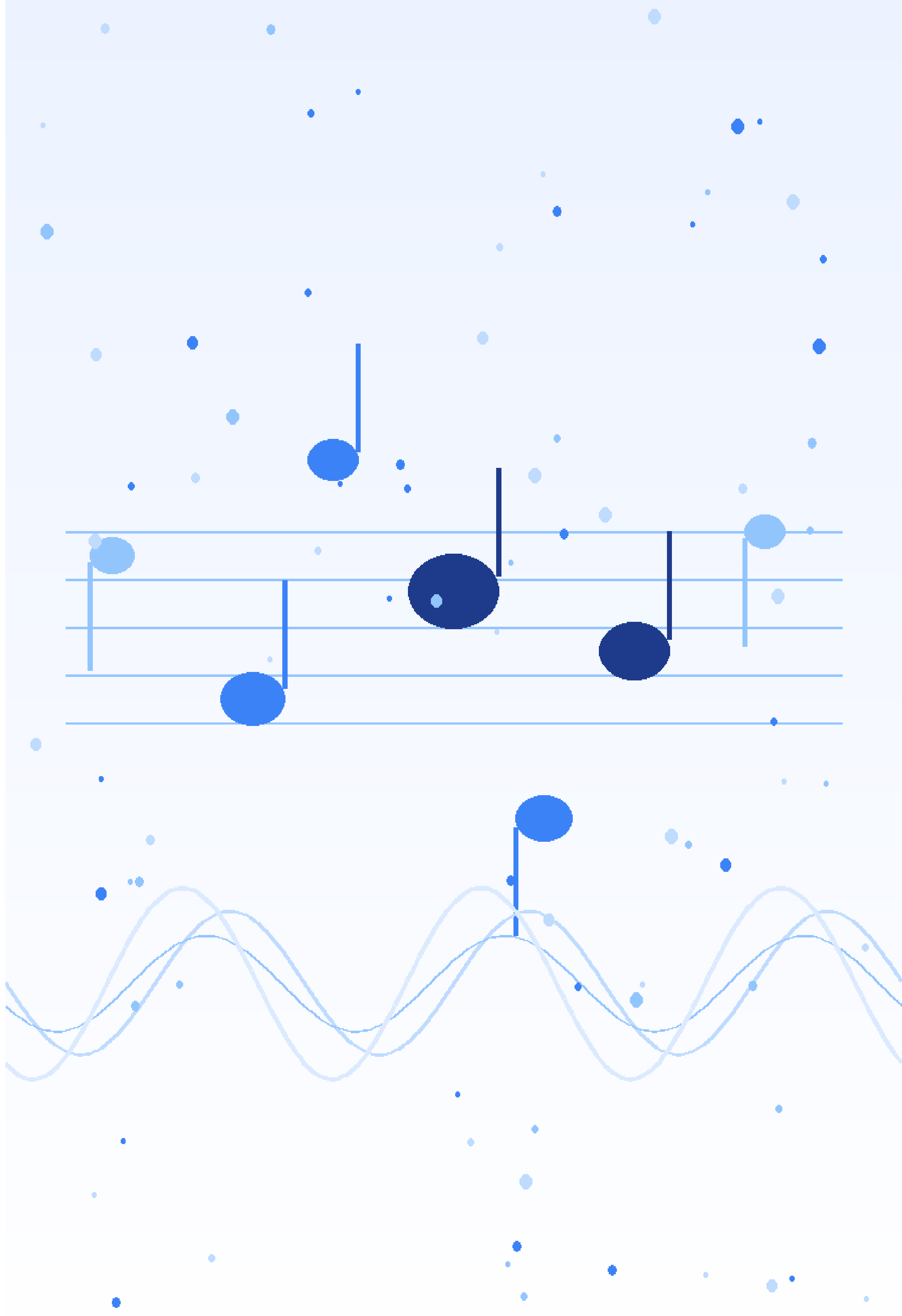
DEPERTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

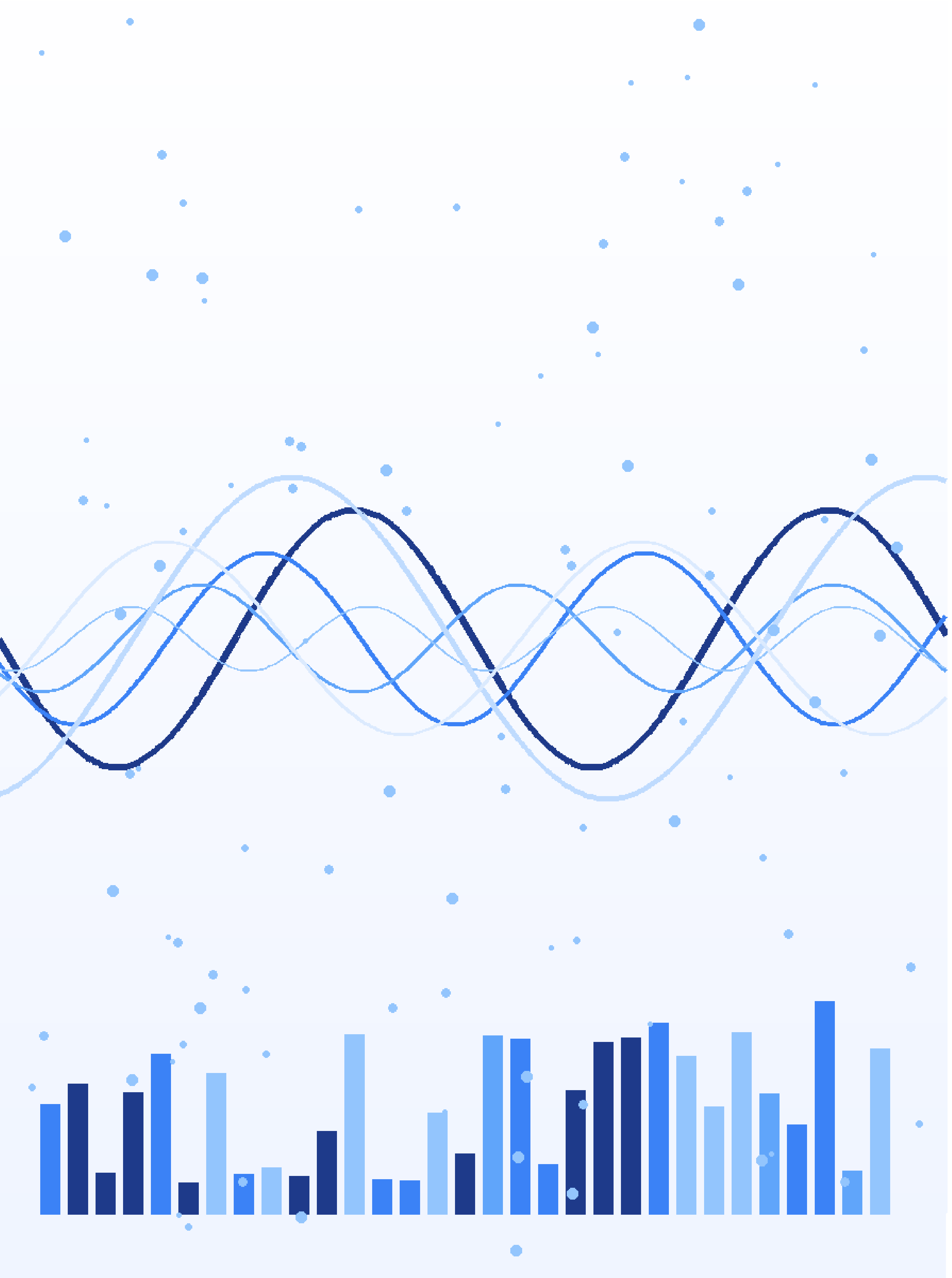
FINAL YEAR PROJECT

AI-Powered Music Generation Platform using Diffusion Transformers and Serverless GPU Infrastructure

ACADEMY OF TECHNOLOGY

DEPERTMENT OF ECE





Project Team

Team Members

1. Anit Sarkar (Roll-34)
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4. Anindita Mazumdar (Roll-32)
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Project Guide

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Institution

Academy Of Technology,

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OBJECTIVE OF THE PROJECT

01 Develop an end-to-end AI-powered music generation platform

02 Automate lyrics, music, and thumbnail generation

03 Reduce music production time from hours to minutes

04 Build a scalable backend using serverless GPU infrastructure

05 Provide a user-friendly web interface for non-technical users

06 Implement secure authentication and user management

07 Store and manage generated media efficiently using cloud infrastructure

SUPPORT MULTIPLE GENERATION MODES:

- Prompt to Lyrics, Music and Thumbnail Generation
- Custom Lyrics to music and Thumbnail
- Only Instrumental Mode
- Multi-Lingual (Bengali/Hindi/English) Music Generation

MOTIVATION OF THE PROJECT

TRADITIONAL BARRIERS

- Technical expertise required
- Expensive software/tools
- Significant production time

WHO'S AFFECTED

- Content creators
- Beginners
- Non-Technical Users
- Small media teams

EXISTING SYSTEMS ARE:

- Closed-source
- Difficult to customize
- Limited in flexibility

OUR SOLUTION GOAL

Build a scalable and user-friendly AI-powered platform capable of generating complete music tracks from simple user prompts while demonstrating practical integration of modern AI infrastructure.

METHODOLOGY AND DESIGN

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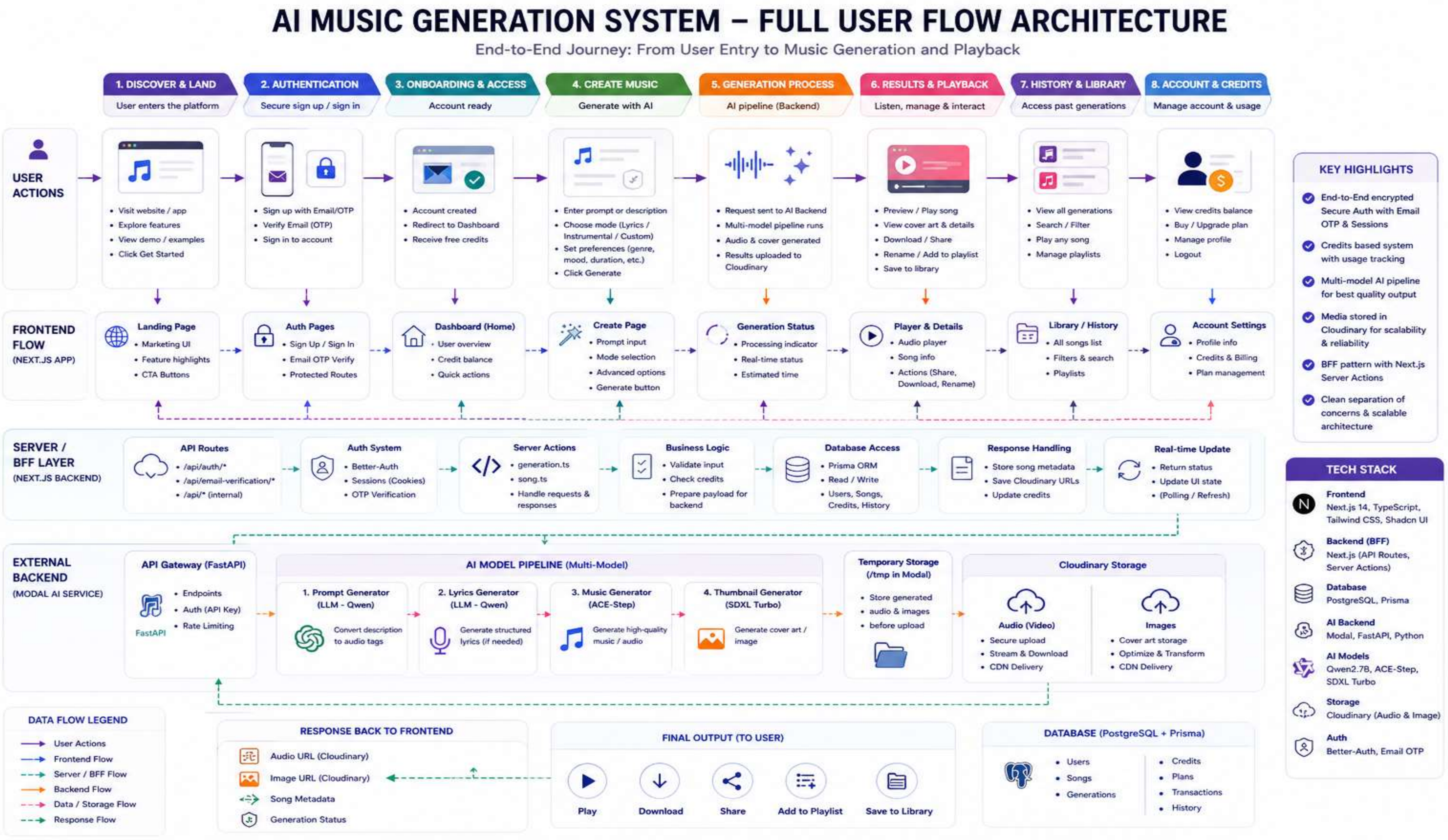

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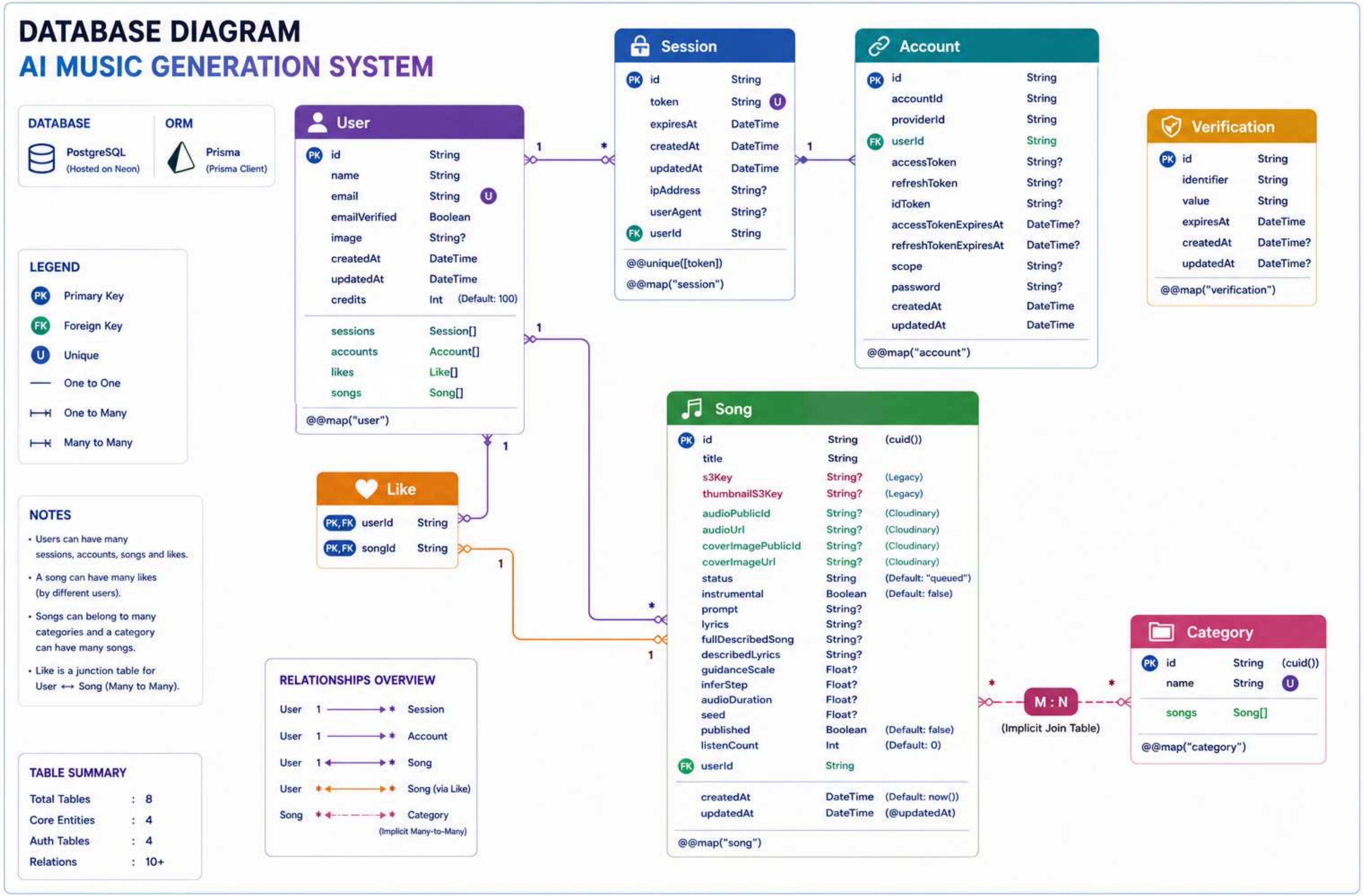

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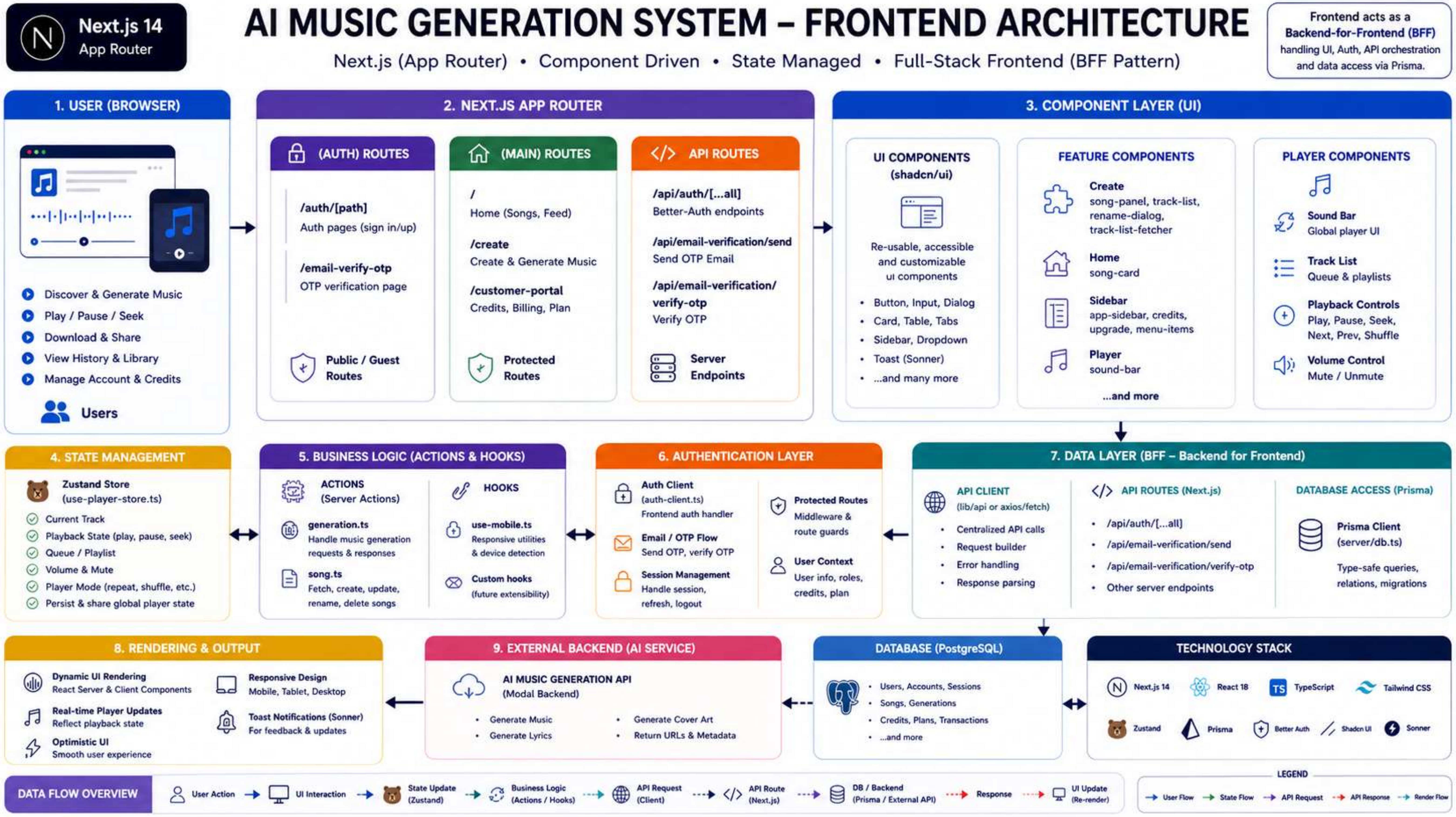

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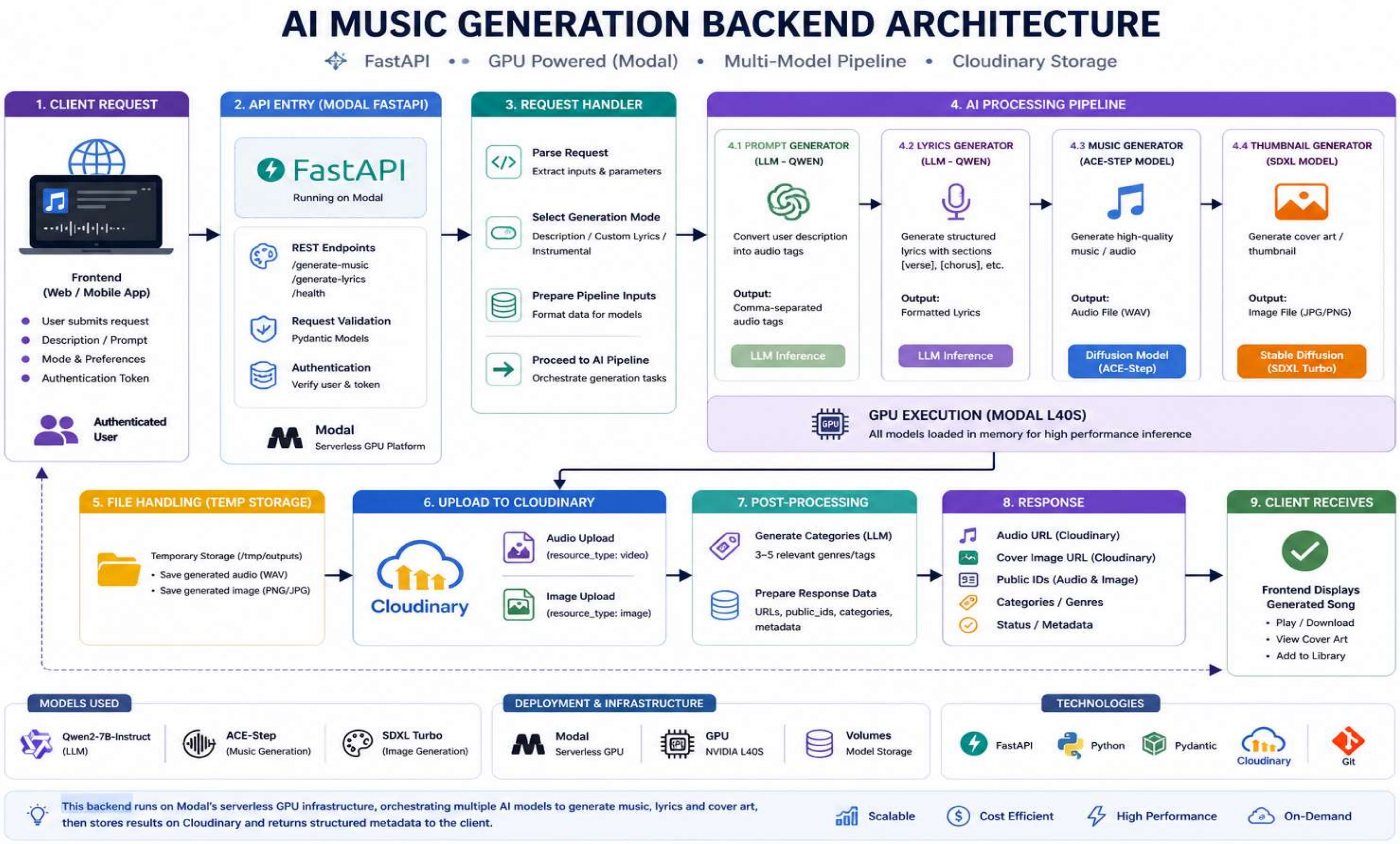

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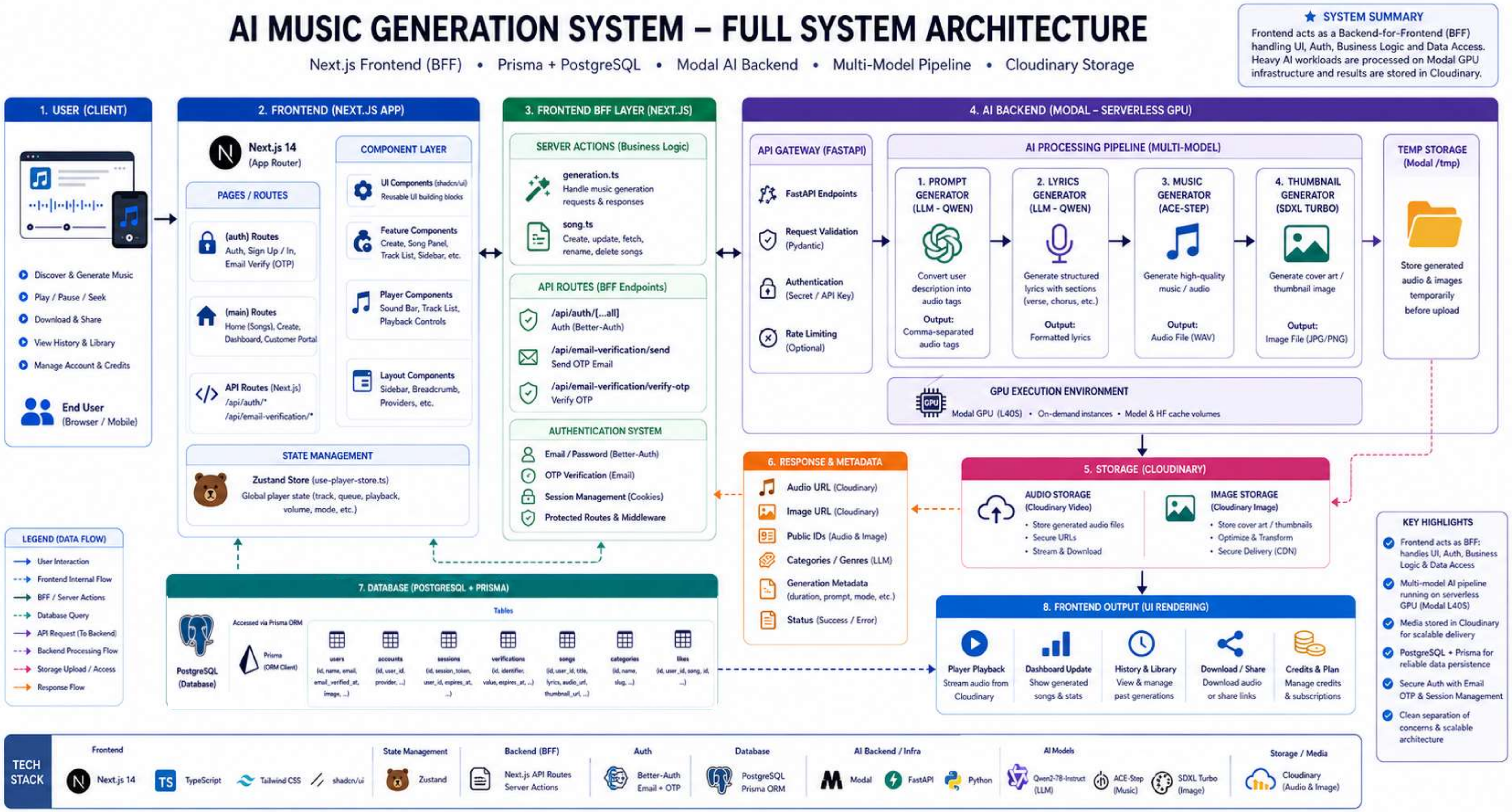

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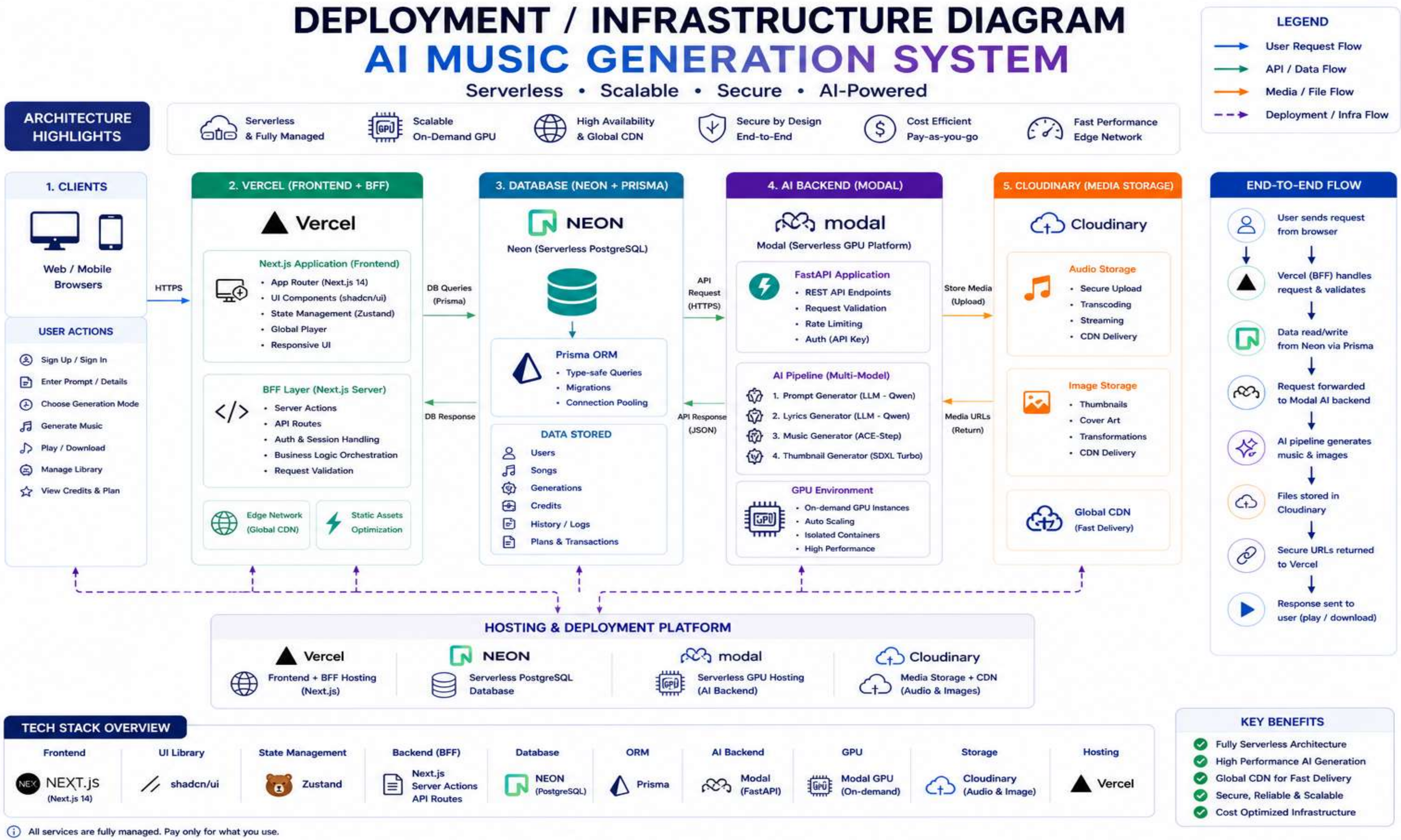

Implementation/ProgressAnd Results/analysis

IMPLIMENTATION/PROGRESS:

- User Authentication and OTP/Link Verification
- Dashboard Creation
- User input field

RESULT AND ANALYSIS:

- User can Sign Up
- User can Sign In
- User can give prompt or describe their song

CONCLUSION

01

This project presents a complete AI-based music generation system that integrates frontend interaction, backend processing, and multi-model AI pipelines into a unified platform.

02

The system successfully automates the process of music creation, reducing manual effort and production time while maintaining usability and scalability. The use of a Backend-for-Frontend architecture, serverless GPU computing, and cloud-based storage ensures efficient performance and modular design.

03

The project demonstrates how AI can be effectively applied in creative domains to build practical and scalable solutions. With further improvements in model quality and system optimization, this platform has the potential to evolve into a fully production-ready system for real-world applications.

FUTURE SCOPE

PLANNED IMPROVEMENTS

1. Bengali and Hindi Lyrics Generation
2. English Music Generation
3. Bengali and Hindi Music Generation
4. Thumbnail generation
5. Image to Lyrics, Music and Thumbnail Generation
6. Cloud Media Storage for Music audio and Thumbnails Saving and Postgres SQL for metadata
7. Media Delivery and Playback in User Side Dashboard
8. Deployment

LONG-TERM SCOPE

→ A production-ready SaaS platform

→ A creator tool for digital media industries

→ An API service for AI-powered music generation

→ A scalable infrastructure for AI creative applications

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Thank you